CSCI 788 — PROJECT SPECIFICATION (REVISED V2, OCT 30 2026)

Grounded Multi-Agent RAG for Healthcare Workflow Automation

Verifiable, policy-grounded decision support for multi-agent LLM systems

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WHY THIS MATTERS

Bigger Picture & Connections

AI is entering high-stakes healthcare decisions

Prior authorization, claims review, and care management are being automated with LLMs. A wrong or unsupported decision doesn't just cost accuracy — it can delay patient care or trigger compliance failures.

Multi-agent RAG pipelines lack verifiable grounding

Multi-agent systems split reasoning across specialized agents, but most implementations don't verify that each agent's output is actually traceable to the retrieved source document — grounding is assumed, not measured.

THE GOAL

What This Project Is About

Build a multi-agent RAG system that produces healthcare workflow decisions where every claim is explicitly traceable to a retrieved source document.

Illustration — a Medicare coverage request

Case facts

Facts of a request for a skilled nursing, home health or inpatient rehabilitation stay

Eligibility Agent

Checks plan and coverage conditions

Medical Necessity Agent

Checks clinical criteria

Documentation Agent

Verifies required documentation

Orchestrator

Cited, structured decision

BACKGROUND AND RELATED WORK

What Has Already Been Done

Area	Summary
Retrieval-Augmented Generation	Grounding LLM outputs in retrieved documents to reduce reliance on parametric memory alone.
Multi-agent orchestration	Frameworks such as LangGraph and AutoGen decompose complex reasoning across cooperating agents. Recent work reports that single agents often match multi-agent systems at equal compute, so a single-agent baseline is required.
Hallucination auditing	Work measuring how faithful LLMs stay to source material, including citation attribution evaluation.
Healthcare NLP & decision support	Criterion-by-criterion checking (e.g., clinical trial matching, medical-necessity checklists) with evidence for each criterion.

The gap this project addresses: few systems combine agent-level task decomposition with explicit, measured grounding evaluation in a healthcare workflow setting — most optimize for task completion or for factuality, not both together.

ATTACK PLAN & MILESTONES

Fall Semester Timeline

REVISED

Sept

Finalize scope, literature review, build case set from public Medicare Appeals Council decisions (22 development, 18 held-out)

Early Oct

Build RAG ingestion pipeline (Medicare manual chapters) and single-agent baseline

Late Oct

Implement multi-agent orchestration in LangGraph; run on local and hosted models

Early Nov

Evaluation framework: groundedness metrics, leak audit, repeat runs, human review of a sample

Mid Nov

Experiments and comparison against the baseline, including a compute-matched comparison; analysis

Early Dec

Final report, poster, and presentation

EXPECTED RESULTS & DELIVERABLES

What Success Looks Like

REVISED DATA

Final deliverable

An end-to-end multi-agent RAG system (LangGraph) producing structured, cited decisions across eligibility, medical necessity, and documentation checks.

Data sources

Policy documents: Medicare Benefit Policy Manual, chapters 1, 6, 7, 8 and 10 (public CMS manual). **Case scenarios:** public, de-identified Medicare Appeals Council decisions, with the Council's outcome as the known-correct label (originally planned as synthetic cases). No real patient or employer data used.

Evaluated across four dimensions on a held-out case set with known outcomes (definitions unchanged):

Groundedness

% of output claims traceable to a retrieved source passage (exact-quote check)

Hallucination Rate

% of claims with no supporting evidence in retrieved text

Decision Accuracy

Agreement with known-correct outcomes on held-out cases

Latency

End-to-end response time vs. a single-agent baseline

CHANGE LOG

What Changed From the Original Specification

Item	Original	Now, and why
Case data	Synthetically generated	Public Appeals Council decisions with real outcomes; avoids self-made data. Limit: facts are the Council's summaries; held-out set is small (18).
Policy source	CMS coverage / prior-authorization guidelines	Medicare Benefit Policy Manual ch. 1, 6, 7, 8, 10. No public prior-authorization requests or decisions exist.
Baseline	Single-agent baseline (latency)	Single-agent baseline compared on all four dimensions, plus a compute-matched comparison.
Models	Not specified	Run through our own LangGraph code on local and hosted models to test whether findings hold across model sizes.
Grounding checks	Traceable to a retrieved passage	Mechanical exact-quote check; whether a quote supports its claim is checked on a human-labeled sample.
Advisor	Prof. Xumin Liu	To be confirmed.